Rivers in the Sky Are Soaking the Pacific Northwest

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It's the wettest time of the year in one of the wettest parts of the country: the Pacific Northwest. This week, heavy rain, high tides, and rising sea levels are hitting the West Coast. Two **atmospheric rivers** will pound parts of Oregon and Washington on Friday and Saturday, after two others already dropped inches of rain on both states earlier this week.

So what is an **atmospheric river**? Think of it as a river in the sky: a narrow band of **water vapor** pushed along by the wind. It dumps rain or snow over a wide area. These rivers can stretch up to 2,000 miles long and 600 miles wide, said F. Martin Ralph, a researcher at UC San Diego's Scripps Institution of **Oceanography**.

Atmospheric rivers are a natural climate pattern. About 50 to 60 percent of California's rain comes from them, said Josh Weiss, a **meteorologist** with the National Oceanic and Atmospheric Administration. The season runs from October to March, and this year it is off to a fast start.

The trouble comes when several rivers arrive one after another. 'What's really problematic is when you get **compounding** impacts from multiple **atmospheric rivers**,' Weiss said flash flooding, urban flooding, and even landslides. Washington's coast could see 20-foot swells this weekend, said Jeff Michalski of the National Weather Service in Seattle.

Right now, the rivers are arriving with El Niño's warm ocean waters and the year's highest tides, called **king tides**, putting coastal and inland waterways at risk of flooding. People may know **atmospheric rivers** by nicknames like Pineapple Express, which describe where the river begins. Forty years of data from the Federal Emergency Management Agency show that 90 percent or more of all flood damage in the Western U.S. comes from **atmospheric rivers**.

Word Treasure

atmospheric river -- A long, narrow band of wet air in the sky that can dump heavy rain or snow over a wide area.

water vapor -- Water in the form of an invisible gas in the air.

meteorologist -- A scientist who studies the weather and predicts what it will do.

Oceanography -- The study of the ocean, its water, and the life and weather connected to it.

compounding -- Happening one after another so the effects pile up and get worse.

king tides -- The highest tides of the year, when ocean water rises farther onto the shore than usual.

Background you need

An atmospheric river is a narrow band of moist air that flows through the sky like a river, carrying huge amounts of water vapor from the ocean toward land.

Wind pushes this band along, and when it reaches land and rises over mountains or meets cooler air, the water vapor cools and falls as rain or snow.

These rivers are a natural part of the weather, and they bring much of the rain that fills rivers and reservoirs along the West Coast.

Problems start when several of them arrive one after another, because the ground is already soaked and the extra water has nowhere to go, which can cause flooding and landslides.

Break it down

LEAD: Heavy rain, high tides, and rising sea levels are hitting the Pacific Northwest during the wettest time of the year.

KEY EVENT: Two atmospheric rivers will pound parts of Oregon and Washington on Friday and Saturday, after two others already dropped inches of rain on both states earlier this week.

+- WHAT: An atmospheric river is a narrow band of water vapor pushed along by the wind that dumps rain or snow over a wide area.

+- SIZE: Atmospheric rivers can stretch up to 2,000 miles long and 600 miles wide.

+- WHEN: The season runs from October to March, and this year it is off to a fast start.

+- RISK: When several rivers arrive one after another, they can cause flash flooding, urban flooding, and even landslides.

+- CONDITIONS: The rivers are arriving with El Niño's warm ocean waters and the year's highest tides, called king tides, putting coastal and inland waterways at risk of flooding.

EVIDENCE: Washington's coast could see 20-foot swells this weekend, and forty years of data from the Federal Emergency Management Agency show that 90 percent or more of all flood damage in the Western U.S. comes from atmospheric rivers.

CONTEXT: People may know atmospheric rivers by nicknames like Pineapple Express, which describe where the river begins.

Why it matters

Atmospheric rivers bring much of the West Coast's rain, but when several arrive at once they can cause flooding and landslides that damage homes and communities. Understanding them helps people prepare for the risks that come with the rain they depend on.

Test what you remember

1. What is an atmospheric river?

(A) A narrow band of water vapor pushed along by the wind
(B) A cloud that stays in one place over a mountain
(C) A wide ocean current that carries warm water along the coast
(D) A river on land that overflows after heavy snow

2. How large can atmospheric rivers get?

(A) Up to 1,000 miles long and 300 miles wide
(B) Up to 500 miles long and 100 miles wide
(C) Up to 200 miles long and 60 miles wide
(D) Up to 2,000 miles long and 600 miles wide

3. About how much of California's rain comes from atmospheric rivers?

(A) About 90 percent
(B) Almost none of it
(C) About 10 to 20 percent
(D) About 50 to 60 percent

4. When does atmospheric river season run?

(A) From October to March
(B) From April to September
(C) Only during the summer
(D) From January to June

5. Why are this week's atmospheric rivers especially likely to cause flooding?

(A) Because no rain fell in the Pacific Northwest earlier in the week
(B) Because several rivers are arriving one after another, along with El Niño's warm ocean waters and the year's highest tides
(C) Because the rivers are bringing cold, dry air from the north
(D) Because the rivers are arriving during the driest time of the year

6. According to forty years of data from the Federal Emergency Management Agency, how much flood damage in the Western U.S. comes from atmospheric rivers?

(A) About half of it
(B) 90 percent or more of it
(C) About two-thirds of it
(D) Less than a quarter of it

Answer key (try the quiz first!): 1: A 2: D 3: D 4: A 5: B 6: B

Questions to think about

MY THOUGHTS (at least 20 words):